

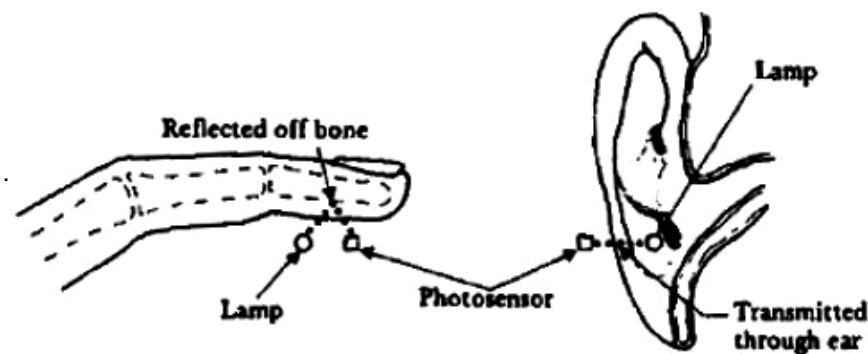
Non-invasive measurement of heart rate and blood hemoglobin concentration through fingertip

I. The Proposed Application : An Introduction

Plethysmographs measure changes in volume. The only method to measure absolute changes in blood volume accurately is by using chamber-plethysmography. The volume change can be converted into blood flow by using $F = dV/dt$, where 'F' is the blood flow per unit time and 'V' is the blood volume. However, in cases where we don't require a very high precision, i.e. we are only interested in the relative volume, for example to see the pulsation rate of the heart; information is in the timing and not in the amplitude or shape of the signal. In that case, we can use electric impedance-plethysmography or photo-plethysmography (PPG), the latter of which is being used here.

PPG is a simple, low cost reliable optical technique that can be used to detect blood parameters by passing Electromagnetic Light rays through the skin. The information about blood parameters is obtained non-invasively contrasted to the painful and inconvenient invasive methods, with measurements generally made at the skin surface whereupon it is easy to record data. During the process, change in volume caused by blood pressure pulse is detected by illuminating the skin from a light emitting diode (LED) and measuring the amount of light either transmitted or reflected through a light detector.

A PPG sensor can be used in reflection mode (for example on the finger) or in transmission mode (for example on the ear) as shown in the figure below. In the proposed analysis, we plan to collect PPG signal via fingertip.



An IR transmitting LED is used to illuminate the subject's fingertip. Variation in collector current for Phototransistor depends on change in blood volume beneath the skin during signal acquisition. Signal recorded is filtered from high frequency noise and unwanted DC component. Output of the signal conditioning circuit is fed to a computer software as input to calculate the heart rate, based on the frequency of the PPG signal obtained (which is a function of blood volume variation) and estimate hemoglobin concentration (through estimation of net IR signal absorbed by the blood matrix).

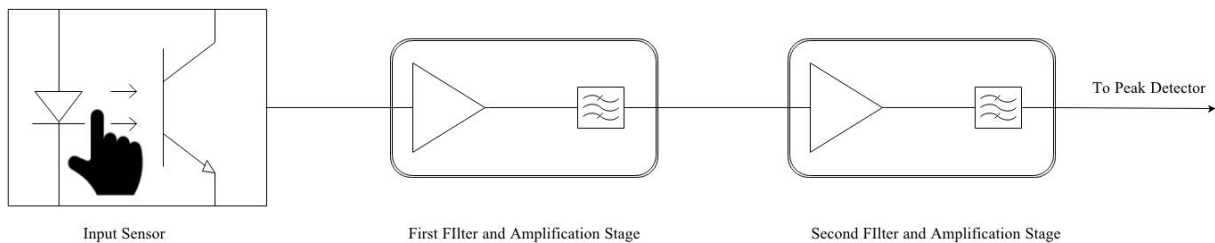
The work proposes to develop a low cost method of estimation of blood parameters non-invasively. The data is collected with the help of an IR LED and Phototransistor pair, and used to generate a PPG signal. The PPG signal is then, after appropriate conditioning, fed to a computer software to compute the blood parameters: heart rate and hemoglobin levels.

II. Description of the functional block diagram & hardware/software components

The project comprises of two parts: Hardware and Software, both of which have been described below.

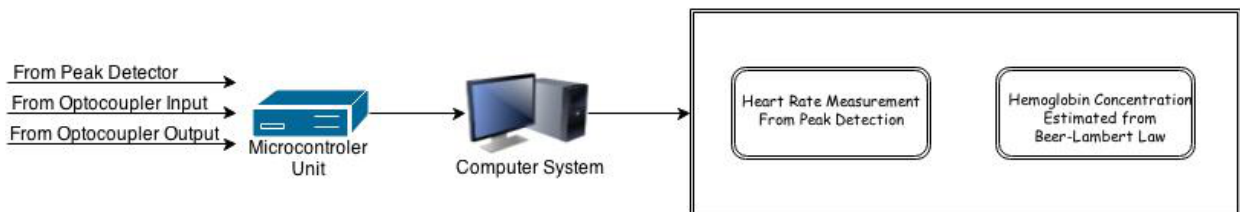
A. Hardware Part :

The Hardware part consists of the PPG signal acquisition circuit. An IR LED acts as the IR emitter, which illuminates the finger and the tissues underneath the epidermis. Due to changes in the blood volume in the finger, the absorption of IR light changes, which leads to variation in the amount of IR signal getting reflected. The change is detected in the phototransistor, varying the amount of collector current through the same. A two stage signal conditioning circuit is used for removing the noise and unwanted DC component from the signal received. It is then fed into the computer for further processing after appropriate Analog-to-Digital conversion through the ADC chip present on the microcontroller and then using serial communication to send the signal to the computer for processing. The microcontroller used here is Arduino Leonardo.



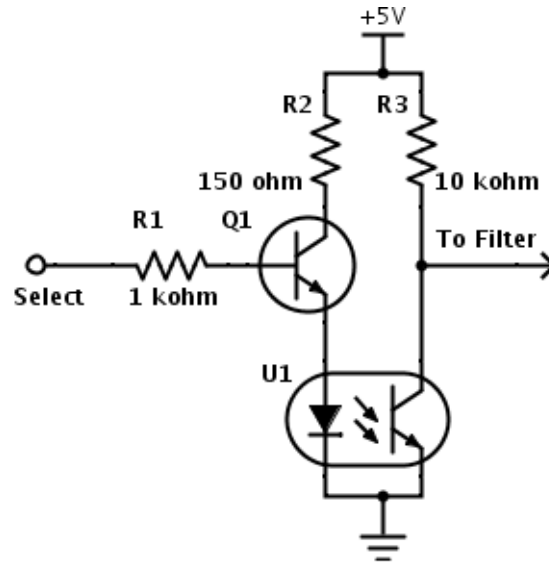
B. Software Part

The Software part consists of codes running on the Arduino IDE and MATLAB®. The enables the setup of a serial communication link between the device and the PC. The PPG signal obtained would be analysed for estimation of Heart Rate and Hemoglobin level. Hemoglobin level is estimated with the help of Beer-Lambert Law, a relation between the intensity of IR radiation absorbed by the blood and concentration of Hemoglobin in the blood.



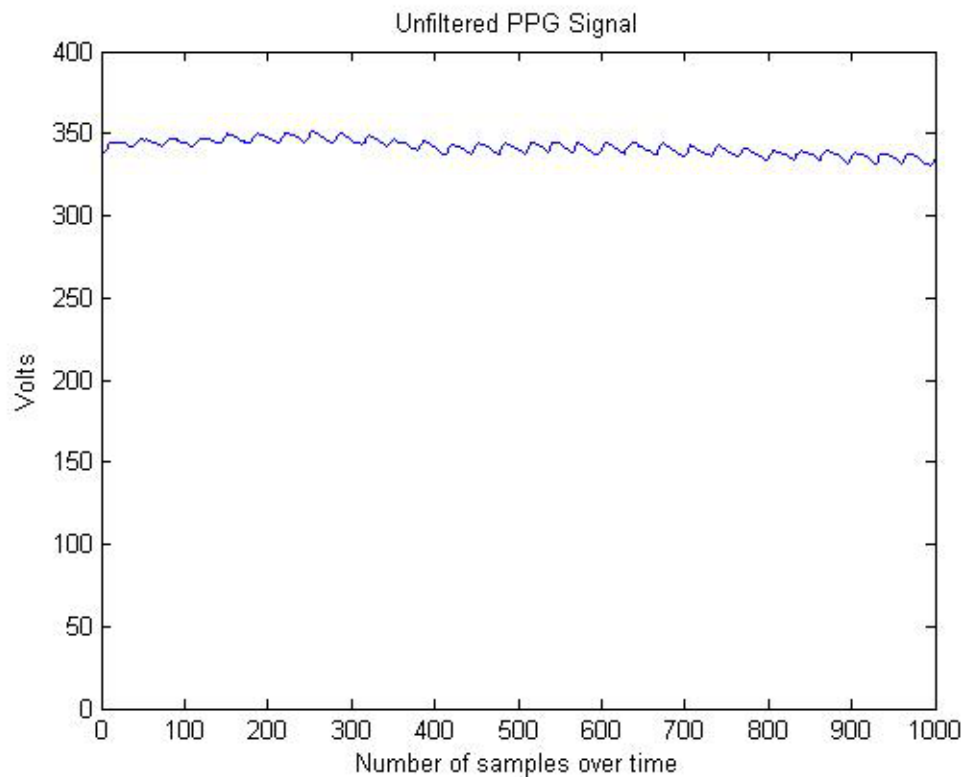
The overall circuit layout as drawn in National Instruments® MultiSim 13.0 is as shown below:

The collector resistance has been taken to be 150 Ω , and the V_{CC} was taken to be 5V. The resistance connecting the phototransistor of the optocoupler to V_{CC} was taken to be 10 k Ω , while the 'Enable' resistance for the *npn* transistor was 1 k Ω .



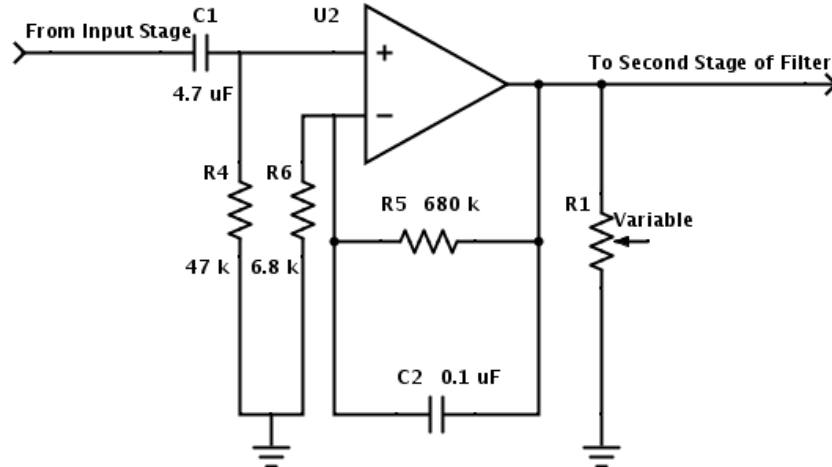
For the sensor, we chose to use the Vishay® TCRT1000, which is a reflective optical sensor with photo-transistor output. It has a peak operating distance of 1mm, and typical output current I_C is 0.5 mA. The emitter wavelength is 950 nm, which is clearly in the Infrared Region of the electromagnetic spectrum, and it comes integrated with a daylight blocking filter.

The output of this stage is as shown below:

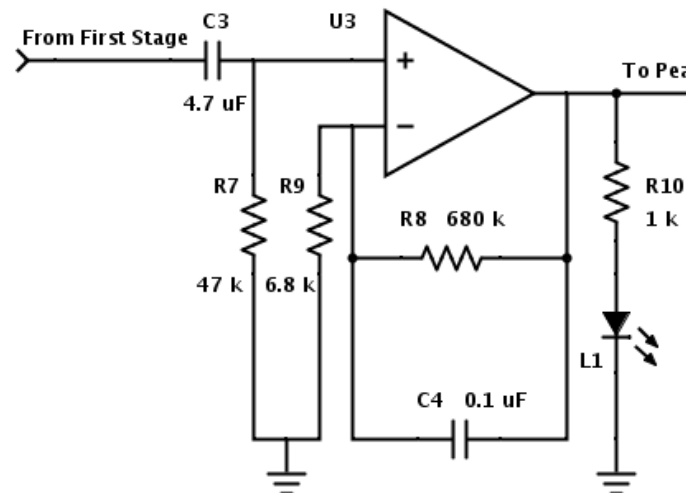


B. Signal Amplification and Filtering Circuit

■ First Stage



■ Second Stage



Design of the signal amplification and filtering circuit:

As mentioned earlier, we want to restrict the PPG signal to have frequencies only between **approximately** 0.5 Hz and 2.5 Hz. Therefore, we needed to choose the resistances and the capacitances accordingly, and they are as shown in the circuit diagrams above.

For the HPF (High Pass Filter): If we choose the capacitance to be $4.7\mu\text{F}$ and the resistance to be $47\text{ k}\Omega$, we get the cutoff frequency of the HPF, f_c as

$$f_c = \frac{1}{2\pi \times C1 \times R4} = \frac{1}{2\pi \times 4.7\mu\text{F} \times 47\text{ k}\Omega} = 0.72\text{ Hz, which can also be taken to be the cutoff frequency.}$$

For the LPF (Low Pass Filter): We can choose the capacitance and the resistance to be $0.1\mu\text{F}$ and $680\text{ k}\Omega$ respectively. In that case, we get the cutoff frequency of the LPF, f_c as

$$f_c = \frac{1}{2\pi \times C \times R} = \frac{1}{2\pi \times 0.1\mu\text{F} \times 680\text{ k}\Omega} = 2.34\text{ Hz, which is quite close to the desired } 2.5\text{ Hz.}$$

Obtaining the desired gain: As mentioned before, the signal which is of interest to us is the AC component, which is quite small in magnitude as compared to the DC offset. Therefore, in addition to the removal of the offset, it is emphatic that the AC component be amplified. We need a **magnification of the order of 10^4** in order to clearly see the signal.

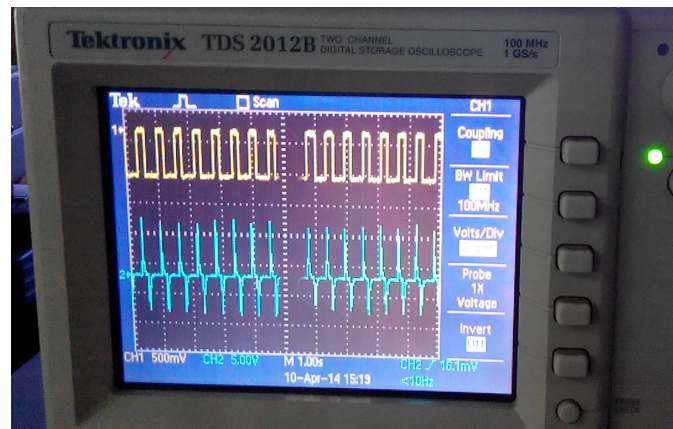
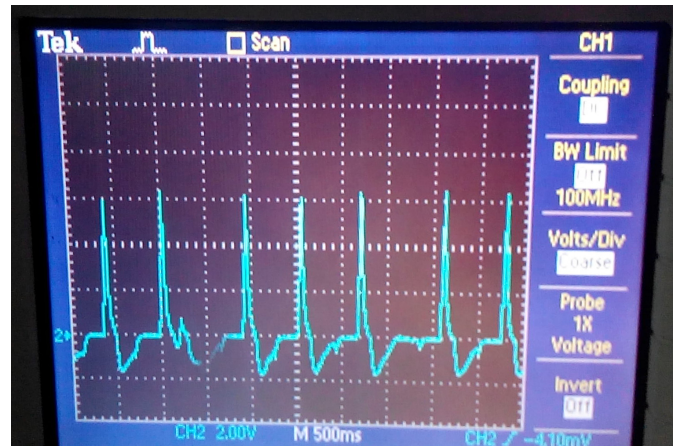
We need to calculate the gain that this circuit provides. Let the gain offered by each stage be A (as both the stages are identical). Thus the overall gain of the circuit would be

$$A_{\text{overall}} = A \times A = A^2$$

$$A = 1 + \frac{R_5}{R_6} = 1 + \frac{680\text{ k}\Omega}{6.8\text{ k}\Omega} = 1 + 100 = 101$$

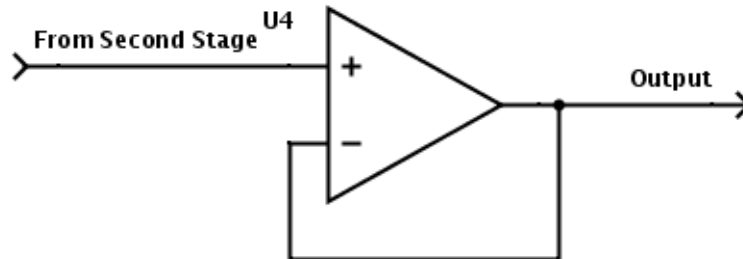
$$\text{So, } A_{\text{overall}} = A^2 = 101^2 = 10201 \approx 10^4$$

The output after these 2 stages are :



C. Peak Detector

We have used LM339 IC in order to make a peak detector. The function of this peak detector is to give a HIGH value whenever the input signal (which is actually the filtered and amplified PPG signal) is at its maximum, which in fact corresponds to a beat. Therefore, in a way, the peak detector circuit gives a HIGH output whenever a beat is detected.



IV. Results

A. Calculating the heart rate

As mentioned in the circuit layout, the peak detector part of the circuit is high whenever the heart beats. As a result, in order to find the heart rate, all we need to do is find the number of peaks in the output and record the time in which those beats were detected. So, we can find the frequency of beats, and then multiply that frequency by 60 in order to obtain the conventional **bpm** representation. This operation has been performed using MATLAB.

B. Calculating the blood hemoglobin level

Beer Lambert Law relates the absorption of light properties to the properties of the material through which light is getting transmitted. In the presented work, we propose to estimate the hemoglobin levels through Beer Lambert Law. The law states that $T = \frac{I}{I_0} = 10^{-\epsilon c l}$

Here I is the intensity of light successfully transmitted through the material,

I_0 is the intensity of light incident on the material,

l is the length through which light travels inside the material,

c is the concentration and

ϵ is the molar absorptivity of the material.

We have taken the following values for these variables :

$$\epsilon = 602.4 \text{ cm}^{-1} \text{ M}^{-1}$$

$$l = 0.6 \text{ cm (thickness of the finger at the point of contact)}$$

$$1 \text{ gm dL}^{-1} = 0.1551 \text{ mM}$$

In this case, we use the input voltage to optocoupler and output voltage from the same as parameters to the Beer Lambert Law. Since we are monitoring the reflected light we use the law as

$$\frac{(I_0 - I_R)}{I_0} = 10^{-\epsilon c l}$$

The input voltage (V_{in}) and the output voltage (V_{out}) are used as substitutes for the incident intensity

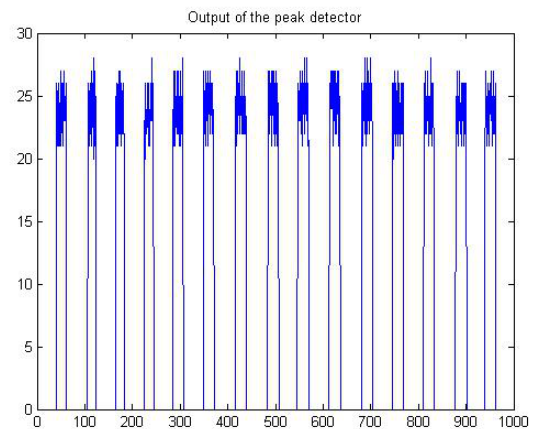
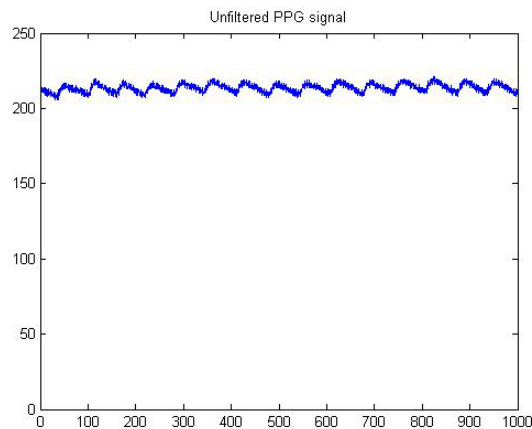
and the reflected intensity respectively due to their linear dependence on the same. Thus we use the following equation in MATLAB to estimate the hemoglobin concentration:

$$\frac{V_{in}-V_{out}}{V_{in}} = 10^{-\epsilon cl}$$

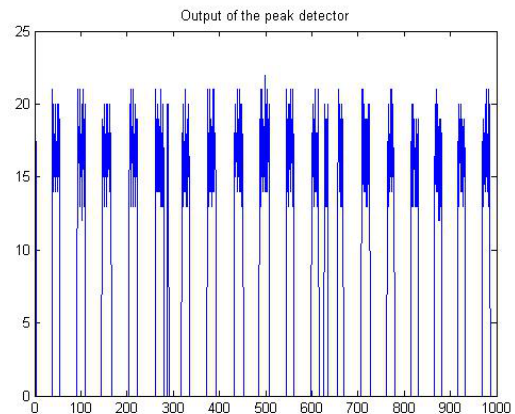
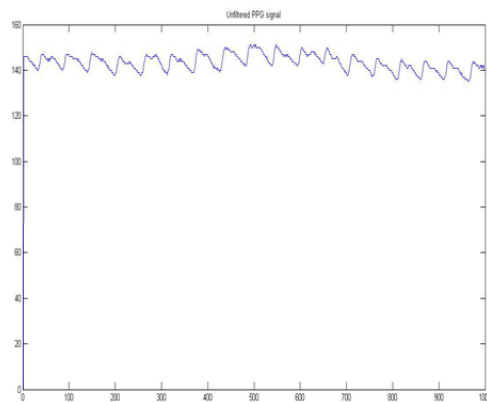
#	Name of subject	Actual Heart Rate (bpm)	Observed Heart Rate (bpm)	Observed Hemoglobin Concentration (gm L ⁻¹)
1.	Kumar Abhishek	78	75	128
2.	Amodh Kant Saxena	92	90	137

C. Plots for the above obtained results

■ Kumar Abhishek



■ Amodh Kant Saxena



V. Summary and Future Work

The project aims to achieve a convenient and comfortable, yet inexpensive way of blood parameter estimation avoiding the inconvenience of blood parameter estimation invasively, by blood extraction. It uses the PPG signal which depends upon the blood volume variation through the finger, using an IR

transmitter-receiver pair. The signal is then amplified and filtered through hardware circuitry and software interface. Analysis is performed on the signal to get the heart rate and blood hemoglobin level.

The PPG signal can help in acquisition of other important blood parameters as well, such as blood pressure and blood glucose by corresponding signal analysis. Future augmentation to the work can also be in the direction to make the process increasingly portable and convenient. This can be done by using the means of wireless connection for sending the data to the computer in place of a physical wired connection. The data would have to be sampled by a microcontroller and sent to the computer after the sampling has been performed.

A much better implementation of this entire setup would be to make it a standalone unit. 'Standalone' means removing the role of a computer, and doing the entire processing on a MCU, probably more powerful than Arduino Leonardo. For example, an Arduino Due/Mega or a Texas Instruments® MSP430 can be used, which have more powerful computational capabilities and an ADC having a higher resolution than the one being currently used, which has 8-bit resolution. This will serve the purpose of being able to use this even in rural areas, where computers might not be so easy to come by. If this is implemented as a standalone unit, the only thing required would be a DC supply.

VI. List of components

#	Item Name	Quantity	Provided by	Price (INR)
1.	Vishay TCRT1000	5	Self	250
2.	Arduino Leonardo	1	Guide	1809
3.	IR LED	1	Self	5
4.	Phototransistor	1	Self	150
5.	npn Transistor	1	Self	-
6.	Red LED	1	Self	-
7.	10 kΩ Potentiometer	1	Self	-
8.	OPA4227 / LM324	1	Self	-
9.	LM339	1	Self	-
10.	Breadboard	1	Self	130
11.	Perforated PCB	1	Self	50
12.	IC Base	2	Self	10
13.	Resistors, Capacitors and Connecting Wires	-	Self	-

VII. References

- **Measurement of Blood Pressure using Photoplethysmography**, M Asif-UI-Hoque, Md Sabbir Ahsan, Bijoy Mohajan, *13th International Conference on Modelling and Simulation, UK, 2011*
- **Photoplethysmography (PPG) System**, Geert Langereis, Version 2, February 2010.
- **Luminous Intensity of an LED as a Function of Input Power**, Sean King, *ISB Journal of Physics*
- **Microelectronic Circuits : Theory and Applications**, Adel S. Sedra, Arun N. Chandorkar, Kenneth C. Smith, *Oxford University Press*
- **Characteristics of Phototransistors** (http://www.johnloomis.org/ece445/topics/egginc/pt_char.html)

Signature of Group Member #1

Signature of Faculty Supervisor

Signature of Group Member #2